



TAGMATIC

AI-powered ticket labelling and theme discovery

Built for Ava Industries · Electronic Medical Records

School for Advanced Digital Technology — Integrated Artificial Intelligence

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A backlog nobody can read



From Ava's own project brief

- Over 1,000 development tickets with no scalable way to analyse or prioritise them.
- Semi-structured free text — every ticket written differently.
- Manual triage is a bottleneck that lowers development velocity.
- Related tickets stay invisible, so the same problem gets solved twice.

1,000+

tickets in the backlog

98

labels in the taxonomy

0

training examples required

Today's agenda



Ludwig

System Architecture



William

Labelling & Clustering



Nehal

AI Chatbot & RAG



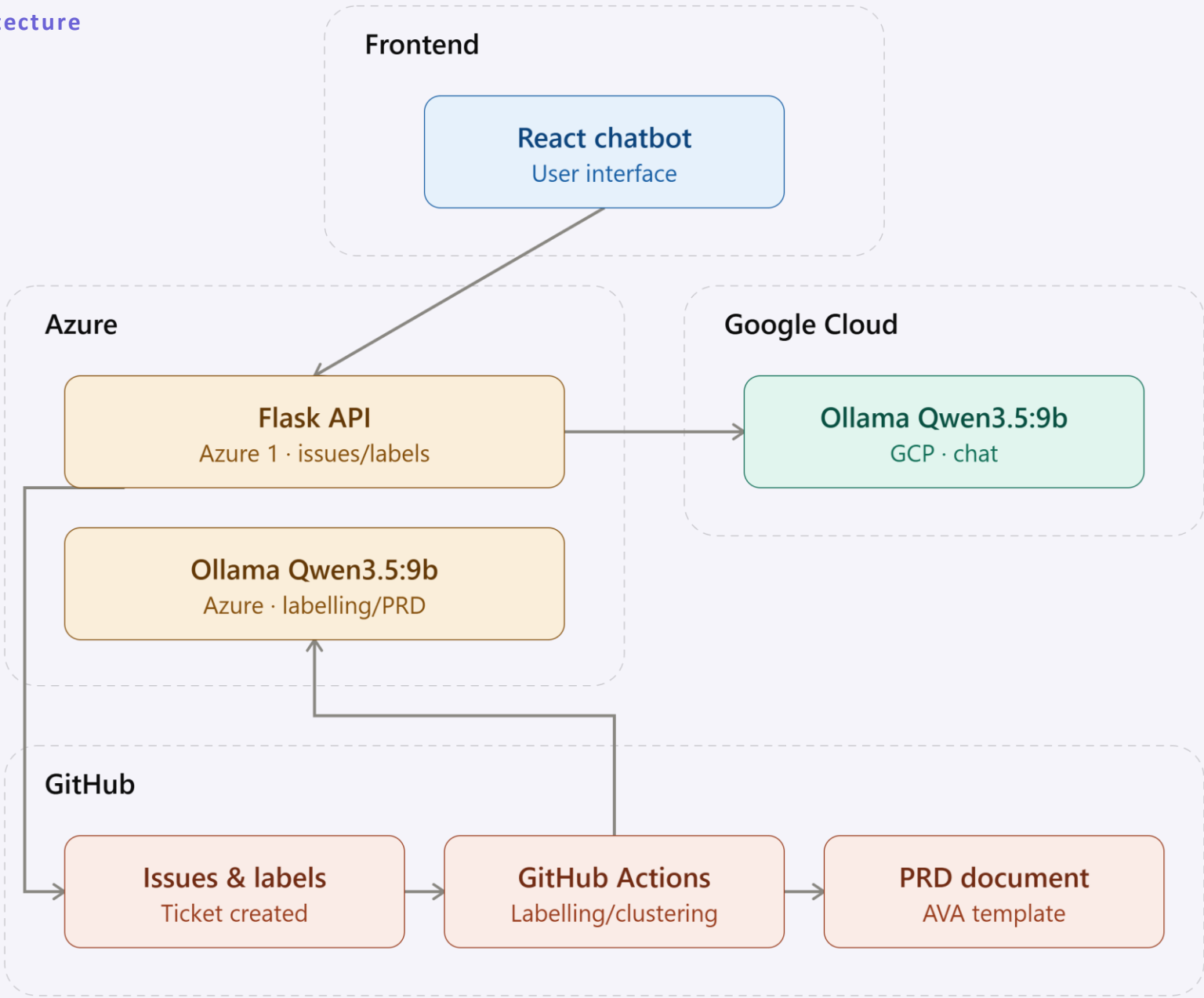
Barbara

PRD Automation



Demo

Everything, end to end



What a labelled ticket looks like

Fake ticket. Real Ava labels.

MADE-UP TICKET AND EXPLANATION

PDF Forms: non-pdfs can be uploaded and used to create corrupt lab requisitions

Upon uploading a new form via the Document Center, the validation logic does not enforce strict file type restrictions...

WHAT THE MODEL SAID

"File-type validation is not enforced on upload, producing unreadable lab requisitions."

0.93 the model grading itself

Labels applied

Type

Bug

exactly one, first

Workflow

Requisitions Management

at most one

Area of EMR

PDF Forms

at most one

General

UI Discrepancies

any number

IF IT CANNOT PICK A TYPE

Review

Needs Human Review

applied alone

If Type is missing, or we had to drop extra Workflow and Area labels, GitHub gets **Needs Human Review** and **nothing else**. A wrong label on a ticket looks finished. A missing one does not.

One request, inside the container



ticket-classifier

uvicorn :8000

1

Receive

FastAPI · Pydantic ·
uvicorn

POST /v1/labelling. Pydantic
TicketRequest checks title
and ticket_id. GitHub Actions
is the client.

>

2

Redact

spaCy · regex

PERSON NER, then email,
phone, URL, clinic names
become [NAME] [EMAIL]
[PHONE] [URL] [ORG].
Prompt only. Raw ticket
stays.

>

3

Constrain

PyYAML · JSON Schema

Load labels from YAML or the
GitHub cache. Only (Label for
AI) ids become schema
enums. qwen cannot invent a
label.

>

4

Generate

httpx · Ollama /api/chat

qwen3.5:9b, think:false,
format set to the schema.
Zero-shot. No examples in
the prompt.

>

5

Enforce

Python

Exactly one Type, at most
one Workflow and Area.
Extras dropped. review_flag
if Type is missing, then Needs
Human Review alone.

↓ HTTP

Ollama, second container

qwen3.5:9b on a T4. Not this
image. HTTP only.

JSON goes back to GitHub Actions. `gh issue edit --add-label` writes the names. This container never talks to GitHub Issues.

Why qwen3.5:9b

Same 2,793 tickets, open and closed, through both sizes.

Same backlog, both sizes

qwen3.5:4b

a good alternative

Per ticket	0.88 s
2,793-ticket run	41 min
100 tickets / day	~1.5 min
Review rate	14%
Self-reported confidence	0.94

qwen3.5:9b

what we shipped

Per ticket	1.29 s
2,793-ticket run	60 min
100 tickets / day	~2 min
Review rate	13%
Self-reported confidence	0.93

THE BILL IS GPU UPTIME

Azure charges the T4 for the hour it is awake, not for which model is loaded. A hundred tickets a day, 9b costs about 40 extra seconds. That does not move the bill, so we took 9b.



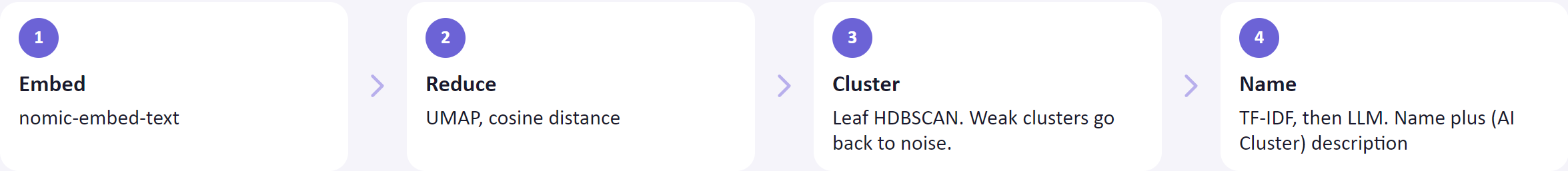
What we tried first

- ModernBERT. Embed the ticket, score cosine similarity against the label descriptions.
- An encoder cannot return a constrained label set, so we could not enforce Type, Workflow, or Area counts.
- No fine-tuning. Almost everything missed the match threshold.

So we switched to a model that returns JSON.

Review rate is whether the JSON was well formed, not whether the labels were right. Confidence is the model grading itself. **We never labelled a gold set, so neither number is accuracy.**

36 themes. Nobody wrote them down first.



MADE-UP NAMES

What a theme looks like

Lab requisition form validation
(AI Cluster) File-type checks failing on lab requisition uploads.

Document export and PDF integrity
(AI Cluster) Corrupt or incomplete PDFs when charts are exported.

Provincial billing claim submission
(AI Cluster) Claims that never reach the provincial payer.

On the backlog	
Tickets in	2,793
Themes out	36
Landed in a theme	63.7%
Left as noise	36.3%

Why not k-means

- k-means needs you to pick the number of themes first.
- Then it stuffs every leftover ticket into one of them.
- HDBSCAN finds the count. One-offs stay noise.

MADE-UP TICKET, LABELS AS THEY APPEAR ON GITHUB

PDF Forms: non-pdfs can be uploaded and used to create corrupt lab requisitions

Type Bug

Workflow Requisitions Management

Area of EMR PDF Forms

General UI Discrepancies

Lab requisition form validation

An AI chatbot grounded in Ava's own knowledge



The model

- Powered by qwen3.5:9b, the same model family behind labelling, run via Ollama.
- Runs as the reasoning engine behind both the chatbot and the PRD drafts.



Retrieval-Augmented Generation

- Retrieves over past tickets and Ava's own documentation before answering.
- Answers “has this come up before?” without a manual search.
- Drafts the ticket for the user through conversation, and accepts file attachments alongside the description.

Why grounding the model actually matters

LLM alone

- Answers from training memory only
- Can sound confident and still be wrong
- No link back to a real source

LLM + RAG (Tagmatic)

- Retrieves real docs & past tickets first
- Answers are traceable to a source
- Stays current as tickets & docs grow



Past tickets act as institutional memory — and this same grounding feeds Barbara's PRD drafts next.

From Issue to draft PRD — automatically



Trigger & gather

A PM runs the Github workflow with an Issue number; it pulls the issue, attachments, and current classification context.



Generate

Drafted by qwen3.5:9b — the same model behind the chatbot — grounded via Nehal's RAG layer over docs.



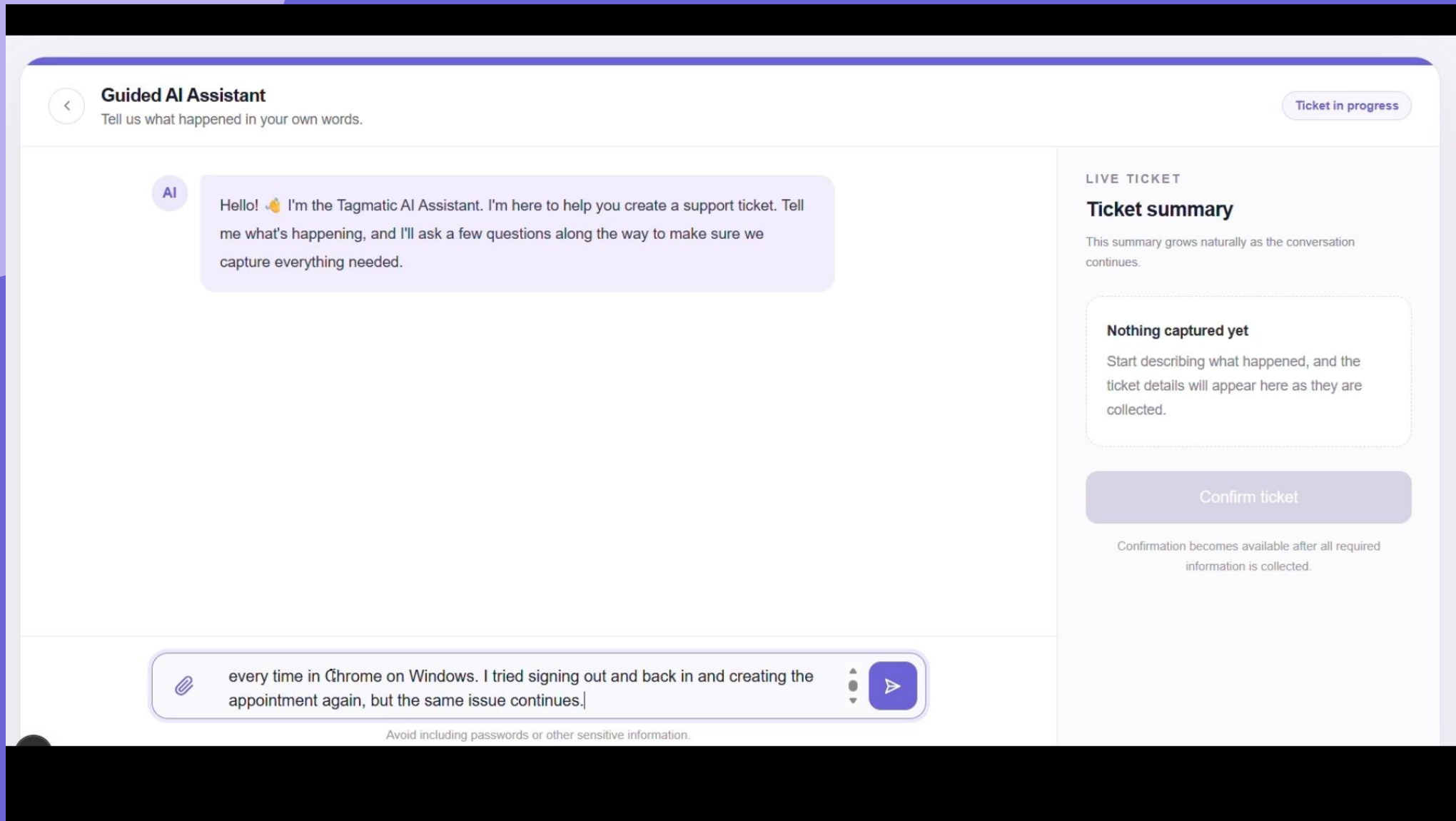
Deliver & flag

Returns JSON with a base64 document, attached to the Issue automatically — also runs as a manual workflow for tickets created elsewhere.



Always a draft, never a shortcut: PRDs are flagged with (Suggested), (Contradiction flagged) and (CRITICAL RISK) markers for PM review.

Let's see it in action



What Tagmatic changes



Every ticket arrives labelled

Type, workflow and EMR area applied on creation — or held for review, never guessed.



The backlog has a shape

Themes emerge from the tickets themselves, rather than from a taxonomy written in advance.



Nothing new to learn

It runs inside GitHub. No second tool, no second source of truth.

What's next

- Feed PM edits on PRD drafts back into retrieval, to improve future grounding.
- Expand the chatbot's RAG sources beyond docs and tickets — release notes, changelogs.
- Broader rollout across more of Ava's ticket queues.



Thank you.

Questions?

Barbara · Ludwig · Nehal · William